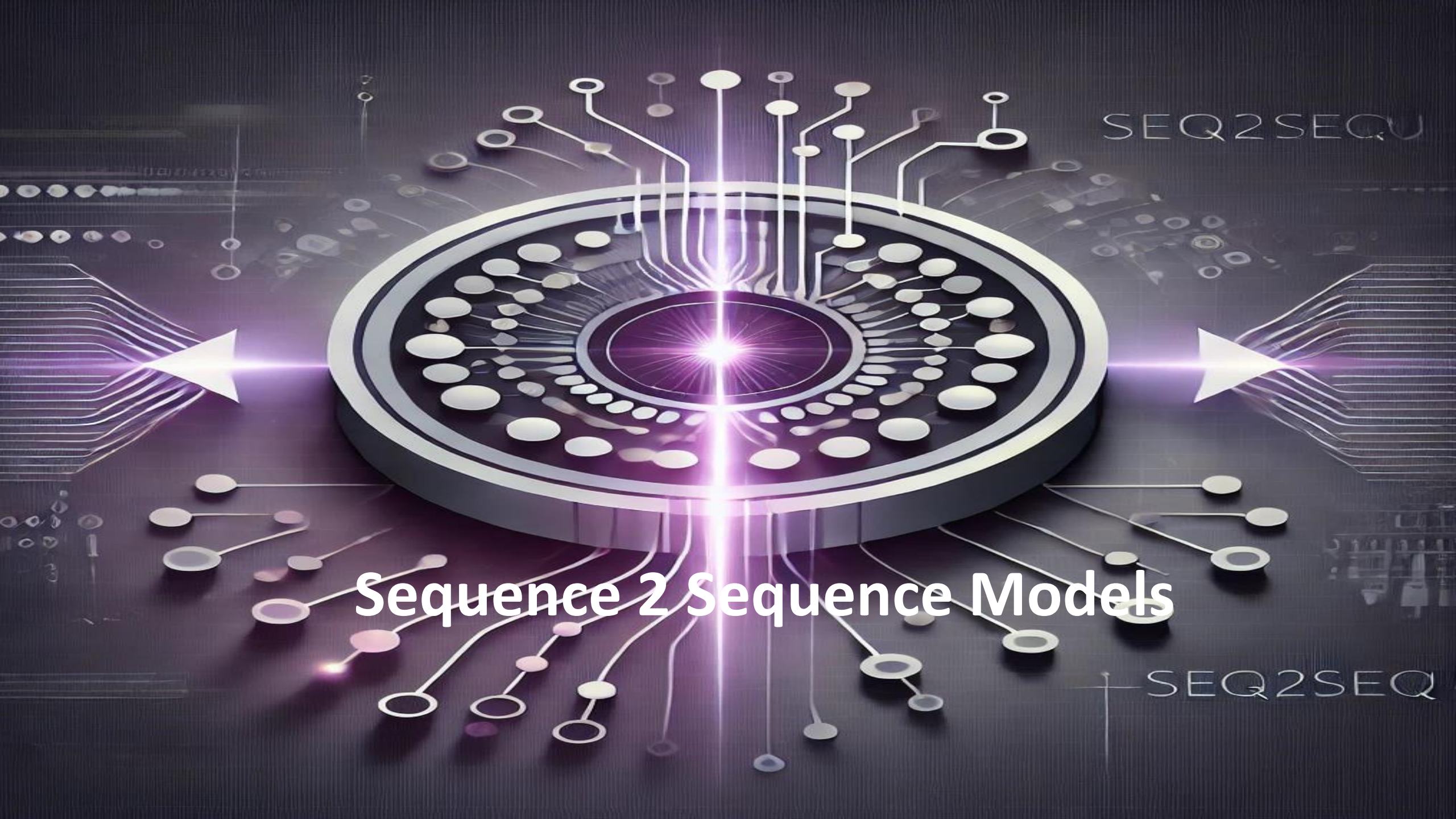


Sequence 2 Sequence Models

A futuristic, glowing purple and white circular interface with a central bright light, surrounded by a network of nodes and lines, symbolizing a complex system or model. The interface has a central bright purple light source with radiating lines. It is surrounded by concentric circles of smaller, glowing nodes. The background is dark with faint, glowing lines and nodes, suggesting a complex network or data flow. Two large, white, arrow-like shapes point towards the center from the left and right sides. The text "Sequence 2 Sequence Models" is overlaid in the center.

SEQ2SEQ

SEQ2SEQ

What is Sequential data ?

- Sequential data is information that has a specific order and where the order matters. Each piece of data in the sequence is related to the ones before and after it, and this order provides context and meaning to the data as a whole.
- Example: “The quick brown fox jumps over the lazy dog.”
- What about: “Fox brown quick the jumps over lazy dog”

Natural language text

- All written language is sequential. The order of words in a sentence or paragraph is essential for conveying meaning and understanding the relationships between ideas.

NLP tasks (prior knowledge)

- Sequence Classification
 - Many-to-one
 - e.g. Sentiment Classification
- Sequence Labeling
 - Many-to-Many
 - e.g POS tagging and Named Entity Recognition (NER)

NLP Models (example)

- Naïve Bayes
- Artificial Neural network
- Recurrent Neural Network

Sentiment Classification



Naïve Bayes, ANNs
RNNs

POS tagging



HMMs
RNNs

Machine Translation



?

Text Summarization



?

Headline Generation

- **Sentence:** "A new study reveals that eating fruits and vegetables daily can reduce the risk of heart disease."

"Fruits and veggies reduce heart
disease risk."

- **Sentence:** "The local community gathered yesterday to discuss plans for the new park development in the city center."

"Community meets for new park
plans."

- **Sentence:** "NASA has successfully launched a new satellite to study the effects of climate change on polar ice caps."

"NASA climate satellite launch."

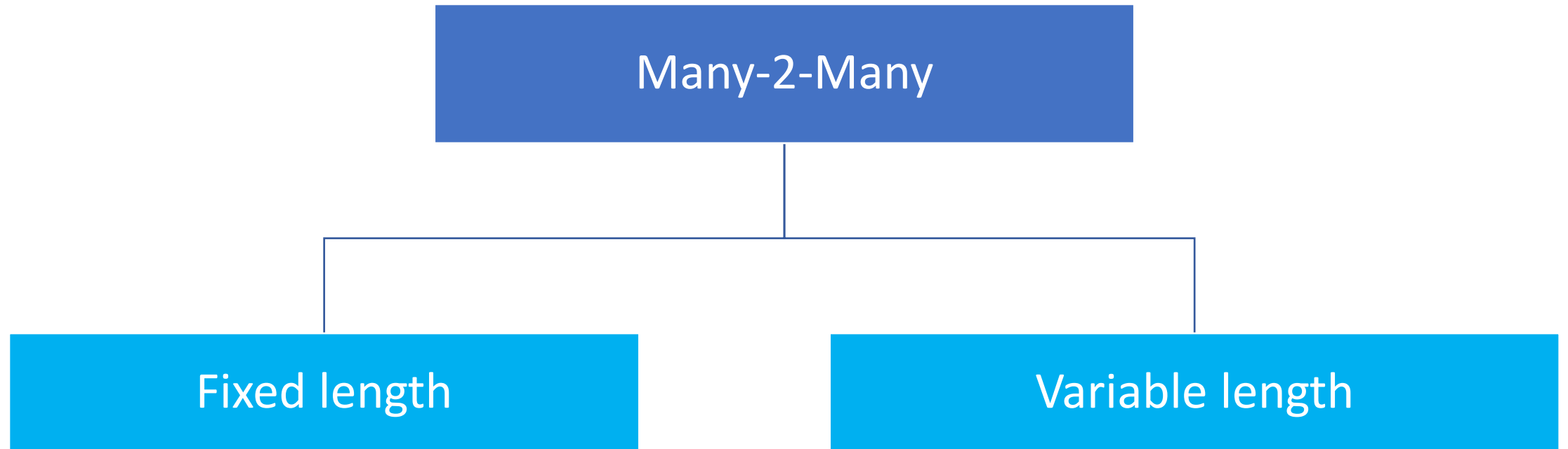
Sequence-to-Sequence/ text generation

- Machine Translation
- Text Summarization
- Question Answering

NLP tasks

- Sequence Classification
 - Many-to-one
- Sequence Labeling
 - Many-to-Many (fixed length output)
- Text generation
 - Many-to-Many (Variable length output)

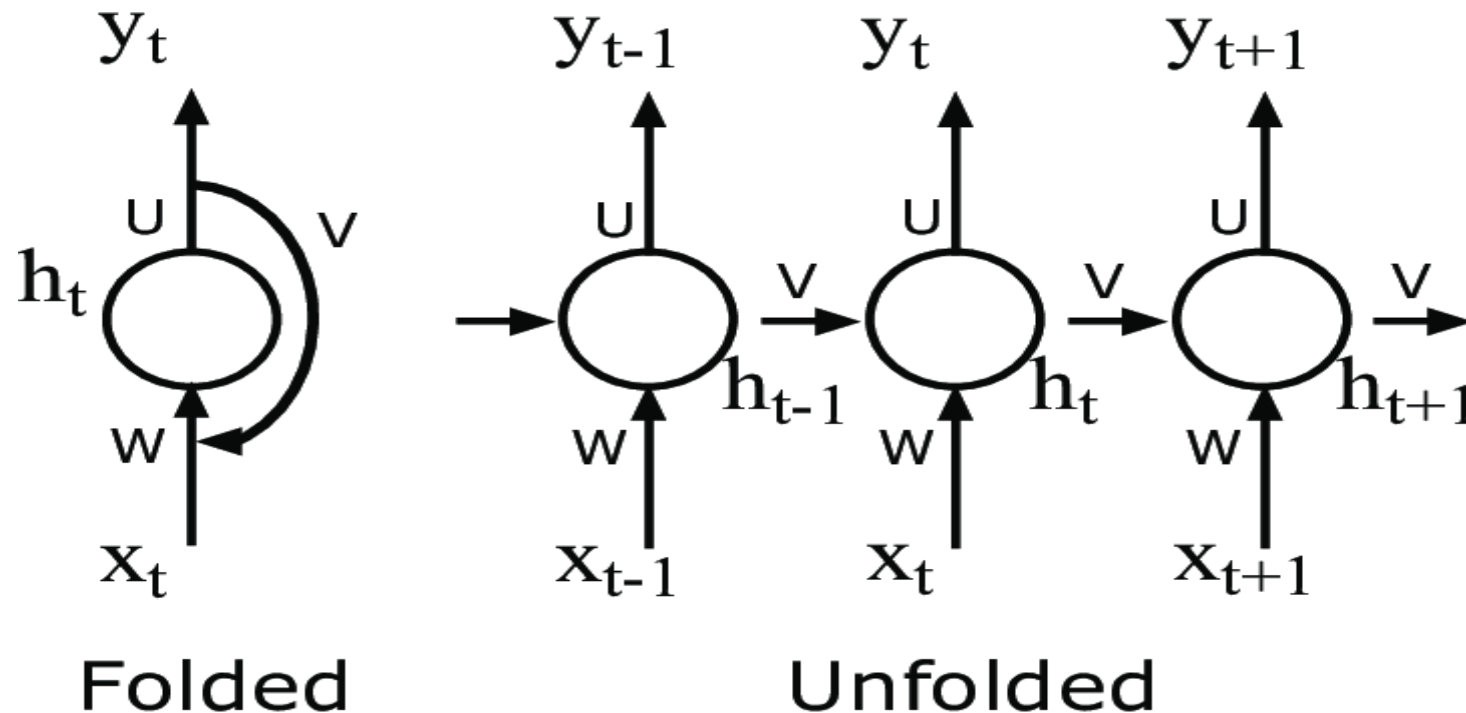
Many-to-Many Tasks (output)



Limitations of RNNs

- Given a sequence of inputs $(x_1, \dots, x_T)$, a standard RNN computes a sequence of outputs $(y_1, \dots, y_T)$ by iterating the following equation:

-



Unfolding

- For example, if the sequence we care about is a sentence of 3 words, the network would be unrolled into a 3-layer neural network, one layer for each word.

- $h_t = g(Vh_{t-1} + Wx_t)$

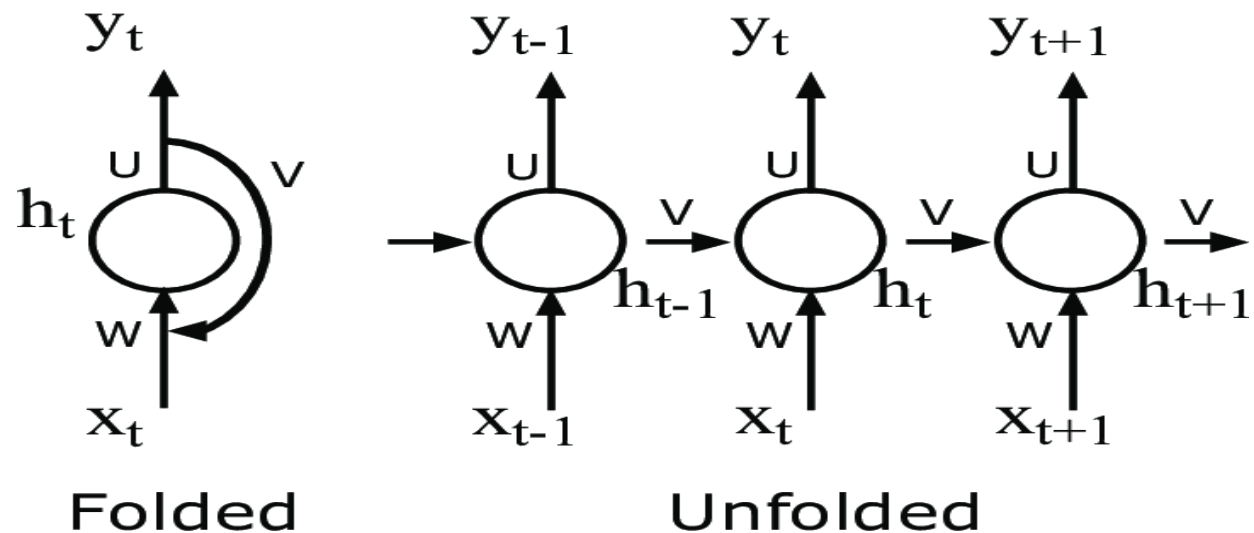
- $y_t = f(Uh_t)$

- $y_t = \text{Softmax}(Uh_t)$

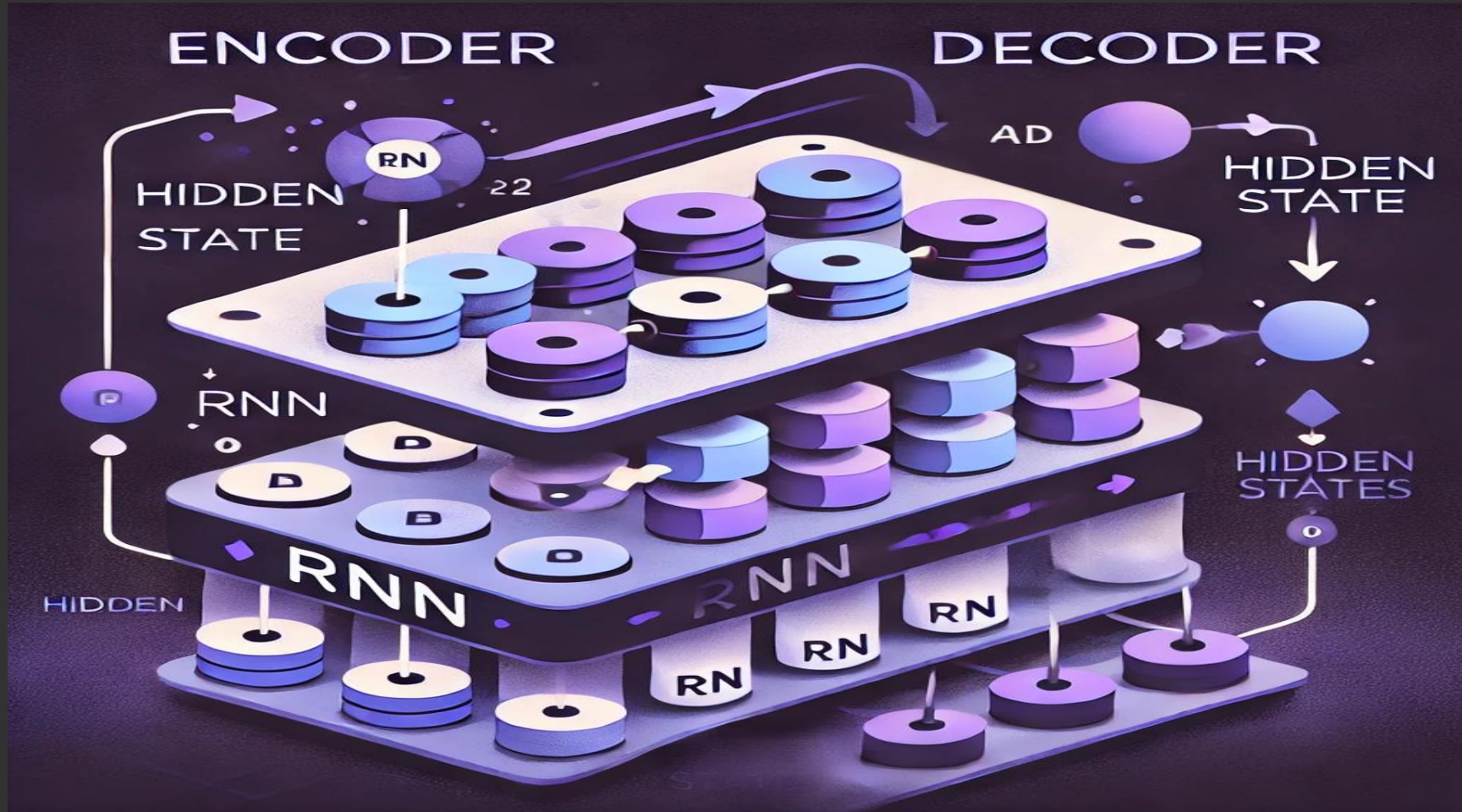
- where h_{t-1} is a *hidden state* that stores the sequence information up to time step $t-1$. In general, the hidden state at any time step t could be computed based on both the current input x_t and the previous hidden state h_{t-1} :

Limitations of RNNs (Continue...)

- The RNN can easily map sequences to sequences whenever the alignment between the inputs the outputs is known ahead of time.
- It doesn't work well for the problems whose input and the output sequences have different length.



Encoder-Decoder Model



Task description

- Input: variable length sequence
- Output: variable length sequence
 - The length of the output is independent of input

- Encoder: Encodes the sequence into fixed length vector, called context vector
- Decoder: Decode the context vector to generate the variable length output



English to Urdu Translation

یہ ماڈل اچھا ترجمہ کے نتائج فراہم کرتا ہے



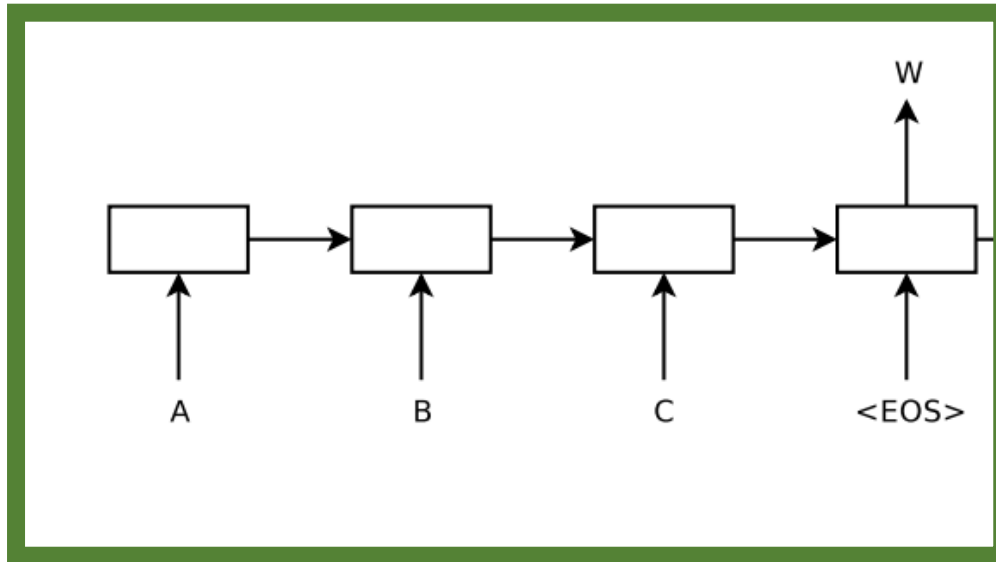
This model gives good translation results

- Encoder: Encoder RNN encodes the sequence into fixed length vector, called context vector
- Decoder: Decode RNN decodes the context vector to generate the variable length output

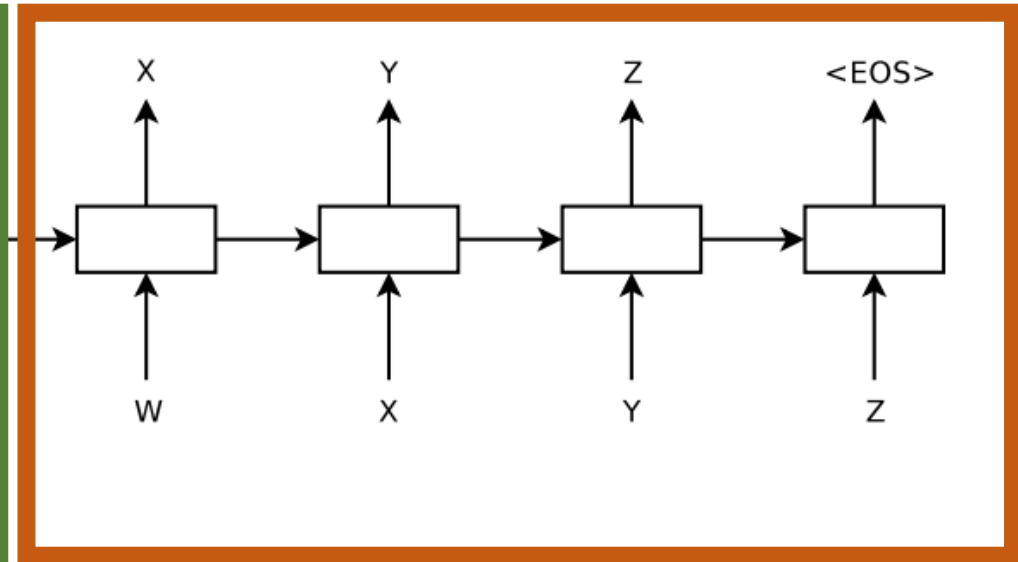


Sequence-2-Sequence Models

Encoder



Decoder



Sequence to Sequence Learning with Neural Networks: <https://arxiv.org/pdf/1409.3215>

- Encoder: Encoder RNN encodes the sequence into fixed length vector, called context vector
- Decoder: Decode RNN decodes the context vector to generate the variable length output



How RNN works?

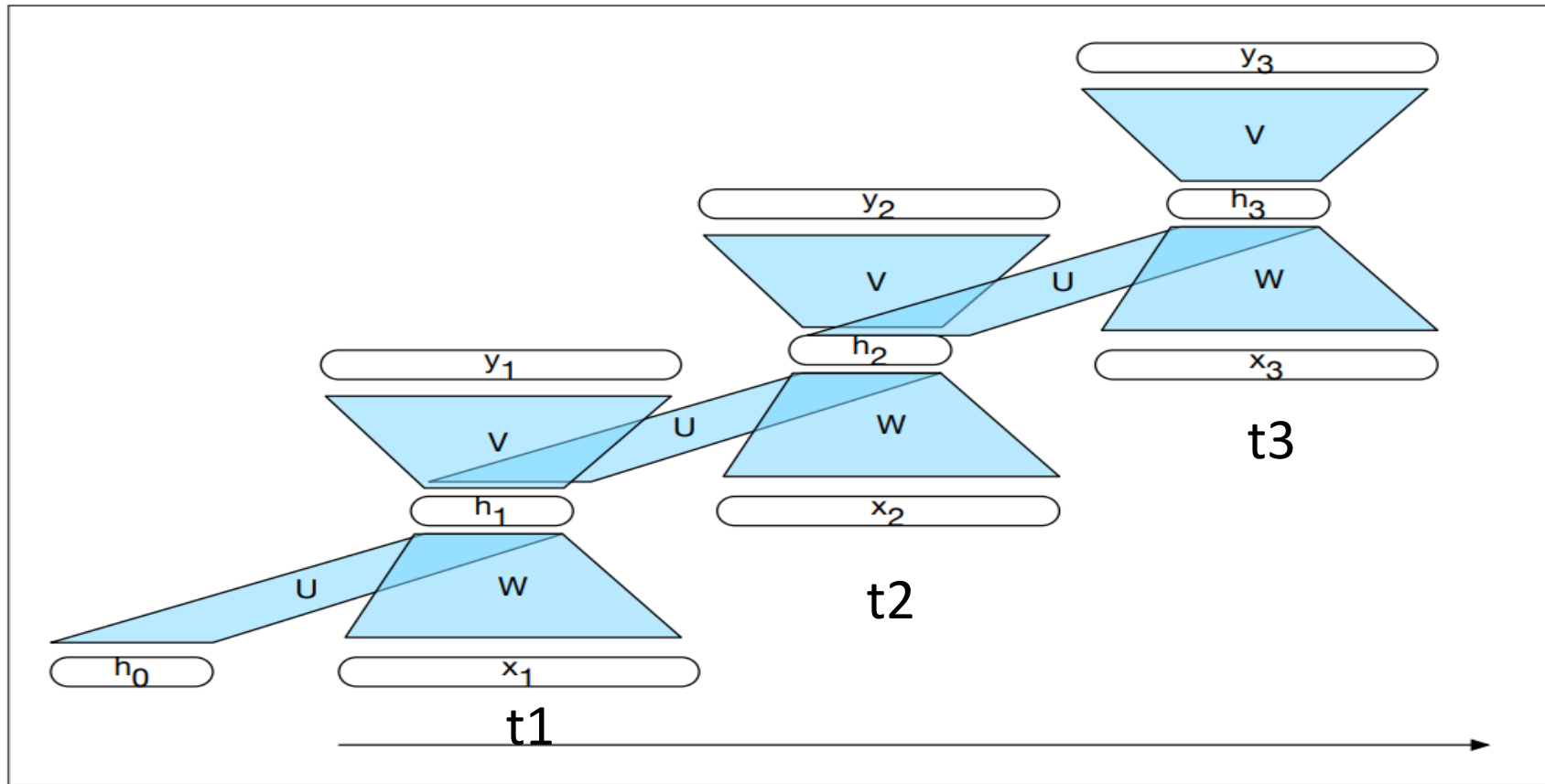


Figure 9.4 A simple recurrent neural network shown unrolled in time. Network layers are copied for each timestep, while the weights U , V and W are shared in common across all timesteps.

Decoder Input?

- Encoder Output
 - The last hidden state of the encoder.
 - $h_t = g(Vh_{t-1} + Wx_t)$

Training (Machine translation)

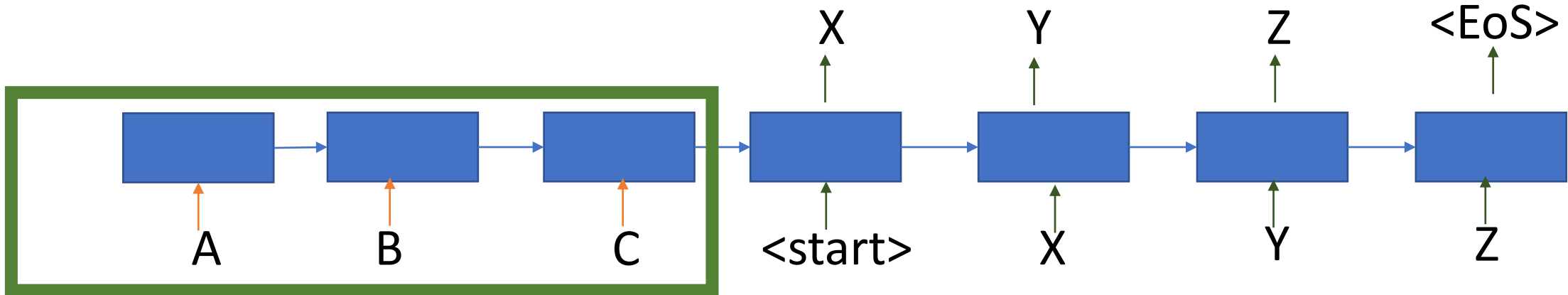
- Dataset: Parallel data set

Input (English)	Output (Urdu)
This model gives good translation results	یہ ماڈل اچھا ترجمہ کے نتائج فراہم کرتا ہے
...	...
...	...

- Tokenization:
 - Encoder: [This, model, gives, good, translation, results]
 - Decoder: [<start>, <EoS>, یہ, ماڈل, اچھا, ترجمہ, کے, نتائج, فراہم, کرتا, ہے]

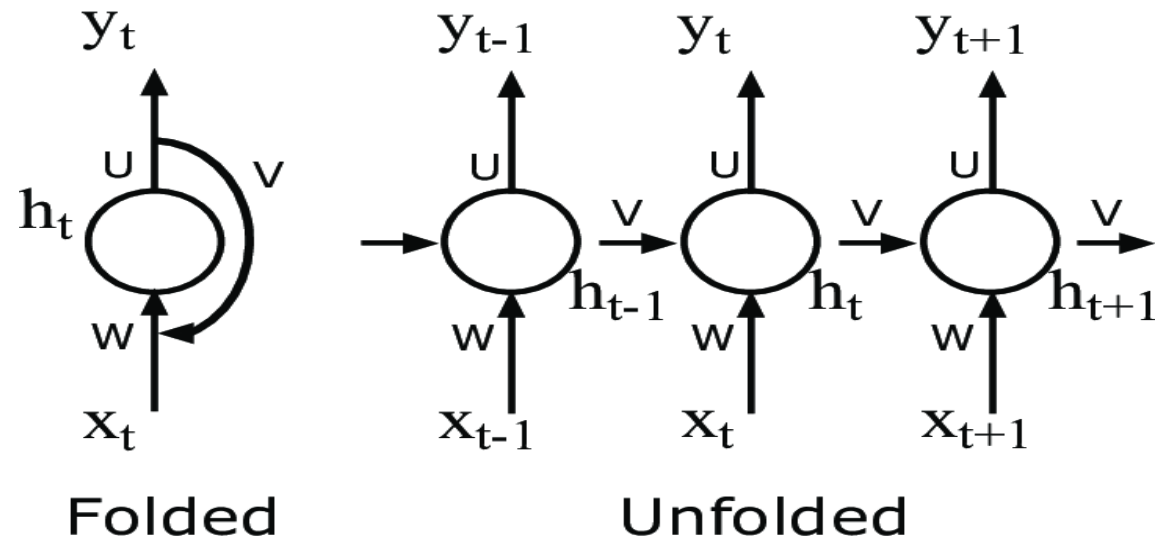
Word Vectors

- Encoder: [This, model, gives, good, translation, results] (6)
- Decoder: [<start>, <EoS>, ماڈل, اچھا, کے, ترجمہ, نتائج, ہوتا, فراہم] (11)
- One-hot-encode vectors dimensions?



RNNs

- Can we share W across encoder and decoder if the vocabulary size is different?



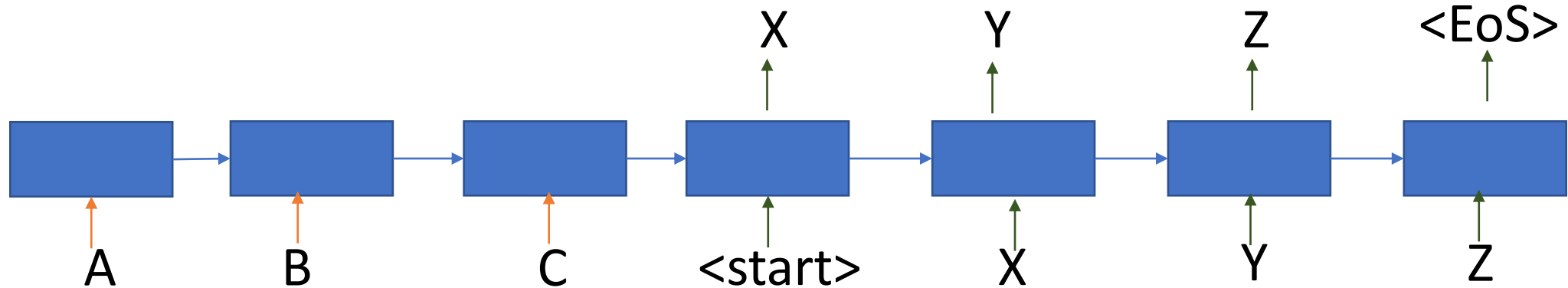
Summary of Weight Usage:

In general, Encoder and decoder use separate sets of weights because they perform different tasks:

- The **encoder** processes the input sequence and summarizes it into a final hidden state.
- The **decoder** takes this final hidden state and generates the output sequence, step by step.
- the input and output spaces can be quite different (e.g., encoding English, decoding Urdu).

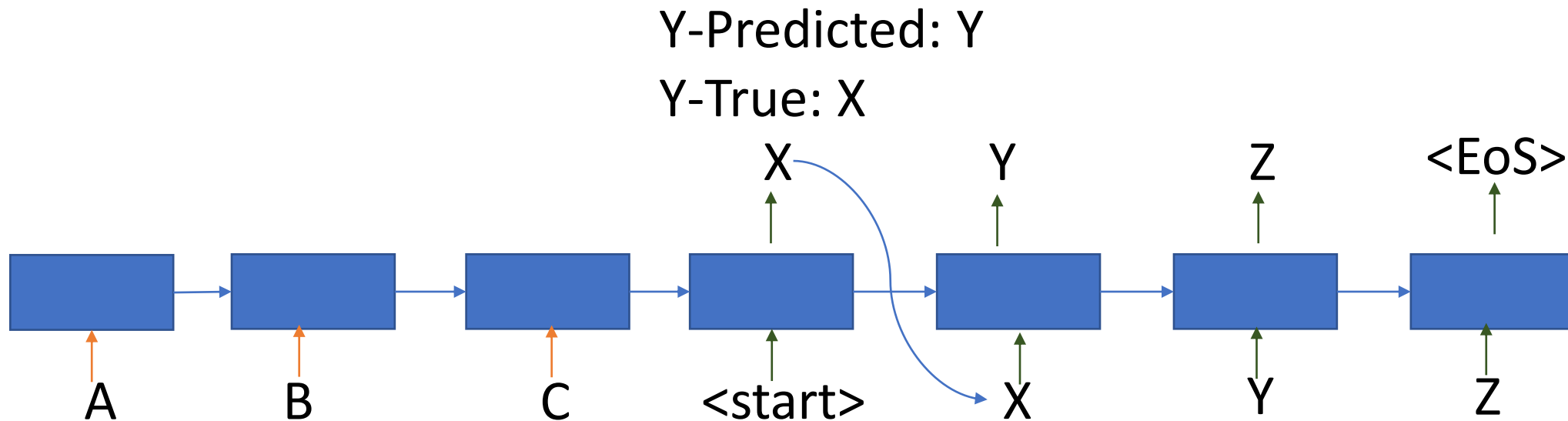
Decoder output layer

- Activation function?
- #classes?



Teacher Forcing (Training technique)

- During training we will condition the output of the decoder on the ground truth even if model makes a wrong prediction.

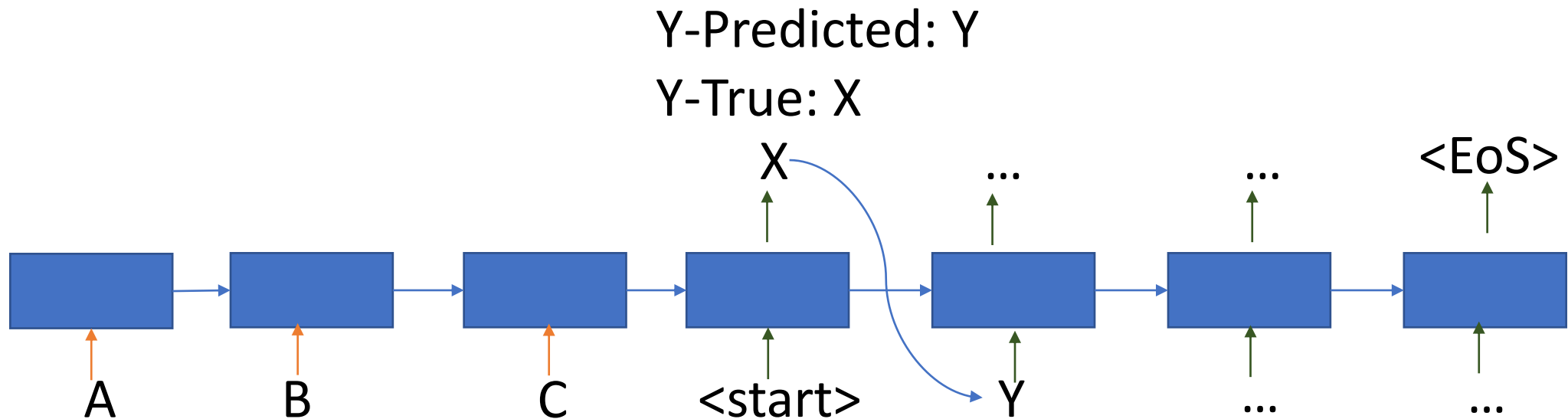


Teacher Forcing

- Teacher forcing is a strategy in which, during the training phase of an RNN-based sequence model, the true target output y_{t-1}^{true} from the training set at time step $t-1$ is used as input for predicting the next output, rather than using the model's previous prediction y_{t-1}^{pred} . This is typically done to improve training stability and convergence speed.

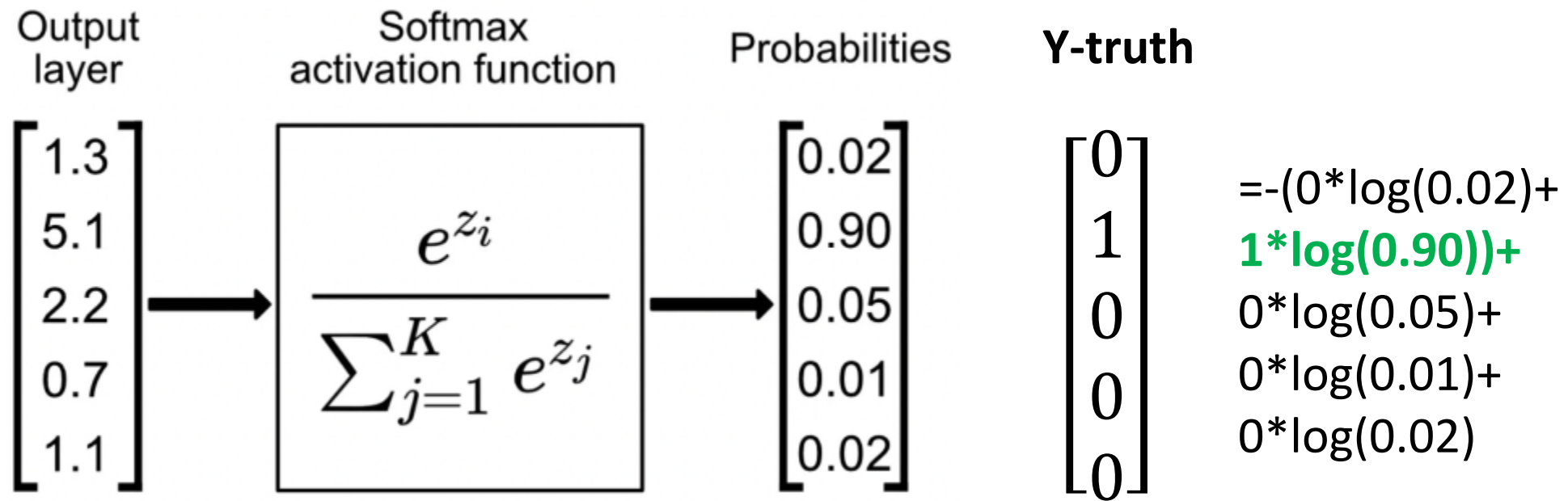
Testing (decoder prediction)

- During testing we will pass the actual output of the time stamp $t-1$ as an input to time stamp t .
- The process will continue till the model predicts $\langle \text{EoS} \rangle$



Cross Entropy Loss

- $H(y, p) = -\sum_i y_i \log(p_i)$



H_0 initialization

- Vector of all zeros

Mathematically:

$$h_0 = \vec{0} \quad (\text{a vector of zeros, typically with the same size as the hidden layer})$$

- Random initialization
- Optimization
- We can learn it during the training process